



SIMATS
ENGINEERING



SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

**GREEN BANKING AND SUSTAINABLE
LOAN MANAGEMNET SYSTEM**
A CAPSTONE PROJECT
**DSA0101 OBJECT ORIENTED PROGRAMMING WITH
C++**

FACULTY

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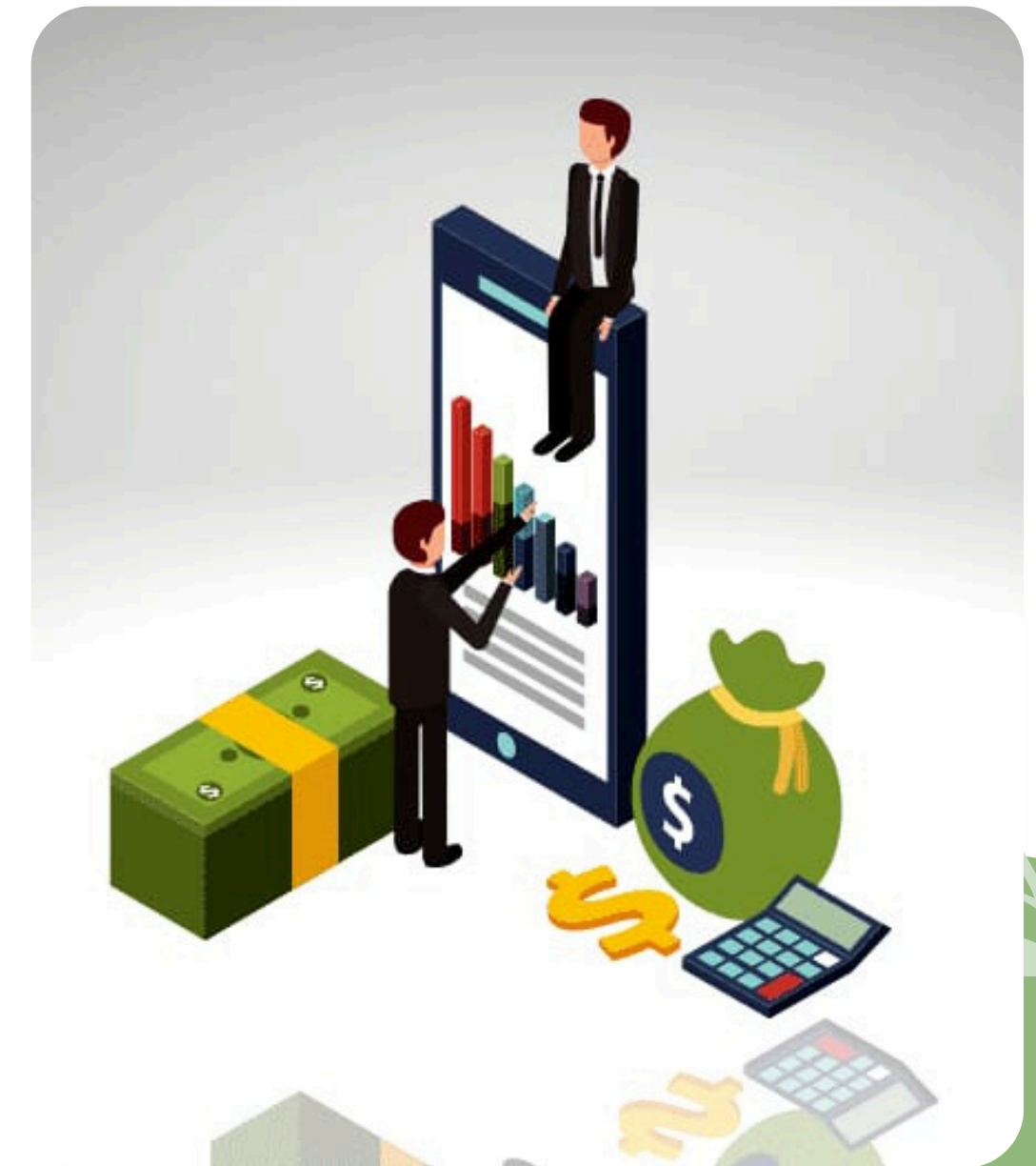
ABSTRACT

Green banking has become an essential strategy for promoting sustainable economic growth while protecting the environment. It encourages banks to adopt environmentally responsible practices in their operations and lending decisions. Banks play a vital role in supporting sustainable development by financing projects that reduce pollution, improve energy efficiency, and promote the use of renewable resources. Green banking also helps financial institutions assess environmental risks before approving loans, ensuring that businesses follow environmental regulations and sustainable practices.



INTRODUCTION

- Green Banking promotes eco-friendly and paperless banking
- Encourages digital banking services and sustainable financial practices
- Supports green projects such as renewable energy and electric vehicles
- Our system is developed using C++
- Manages customer information and sustainable loan applications



LIRERATURE SURVEY

S.No	Author & Year	Methods	Pros	Cons
1	Sahoo & Nayak (2021)	Analysed sustainable finance and credit risk assessment for green loans.	Provides risk assessment factors for sustainable loans.	No automated scoring or approval process.
2	Singh & Sharma (2022)	Studied ESG criteria integration in banking and its effect on loan performance.	Shows positive impact of ESG-based lending on loan performance.	Complex ESG data collection for small borrowers.
3	Patel & Patel (2023)	Proposed a digital model for green loan management in commercial banks.	Demonstrates digital workflow for green loan tracking.	Does not include sustainability awareness tracking.
4	Rathi & Mishra (2024)	Reviewed AI and machine learning techniques for sustainable finance.	AI improves accuracy in credit and sustainability risk prediction.	Requires large datasets and infrastructure.
5	Verma & Gupta (2024)	Developed a framework for green banking with real-time sustainability monitoring.	Real-time monitoring improves accountability and transparency.	Higher implementation cost and complexity.

ISSUES IN THIS EXISITING MODEL

- 1 Banks focus only on financial indicators (profit, collateral, repayment history) and ignore environmental risks
- 2 Traditional loan processing relies heavily on paper, increasing paper waste.
- 3 Lack of awareness about green banking services among customers.
- 4 No standardized framework for evaluating green loans and sustainability.

PROPOSED SOLUTION

1

Develop a Green Banking and Sustainable Loan Management System with a digital, paperless workflow.

3

Use AI-based ESG (environmental social governance) verification and standardized metrics

2

Develop an integrated Green Banking and Green Financing model to achieve SDGs (sustainable development goal).

4

Use paperless online loan application and approval processes

1

**MODULE 1 -
Customer Management Module**

2

**MODULE 2 -
Green Loan Application Module**

3

**MODULE 3 -
Eco-Score Assessment Module**

4

**MODULE 4 -
EMI Calculation Module**

5

**MODULE 5 -
Loan Approval Module**

6

**MODULE 6 -
Transaction & Report Module**



ADVANTAGES

1



PAPERLESS BANKING

Reduces paper usage and minimizes environmental waste by digitizing the loan process.

2



ECO-FRIENDLY LOANS

Promotes and supports sustainable and green projects that help protect the environment.

3



ECO-SCORE SYSTEM

Rates projects based on their environmental impact and encourages better sustainability practices.

4



LOWER INTEREST

Projects with higher eco-score receive interest discounts and better loan benefits.

5



EASY REPAYMENT TRACKING

Maintains repayment history and generates reports for easy tracking and better management.

CONCLUSION

- Applied core OOP concepts to build a real financial system
- Eco-score rewards green projects with lower interest
- Modular design keeps the system maintainable
- Balances sustainability with credit risk management
- Scope to add carbon tracking and mobile access

**GREEN
BANKING**



REFERENCE

- **OECD. (2023). Green finance and banking innovation in emerging markets. Organization for Economic Co-operation and Development**
- **Kumar, R., & Prakash, A. (2023). Assessing the impact of green banking policies on bank performance in India. Journal of Sustainable Finance & Investment, 13(1), 75–90.**
- **Singh, R., & Upadhyay, A. (2022). Green banking as a corporate social responsibility initiative: Indian context. IUP Journal of Bank Management, 21(1), 34–47**
- **RBI. (2022). Guidelines on green deposits and climate risk management. Reserve Bank of India Bulletin.**
- **Bose, S., & Khan, H. (2020). Green banking and environmental sustainability in Indian banks: A critical analysis. Indian Journal of Finance, 14(4), 21–30.**



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Thank You



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